ASSIGNMENT 2 — Q4 — Array Performance Analyzer

Instructions: Save this solution in its own .js file. Run it in Node.js or the browser console.

```
"use strict";
const scores = Array.from({ length: 8 }, () => Math.floor(Math.random() * 71) + 30);
const highest = Math.max(...scores);
const lowest = Math.min(...scores);
const total = scores.reduce((sum, score) => sum + score, 0);
const average = total / scores.length;
const passed = scores.filter(score => score >= 50);
const labels = scores.map((score, i) => `Student ${i + 1}: ${score}`);
console.log("Scores:", scores);
console.log("Student results:", labels);
console.log(`Highest: ${highest}`);
console.log(`Lowest: ${lowest}`);
console.log(`Average: ${average.toFixed(2)}`);
console.log(`Passed (>= 50): ${passed.length} of ${scores.length}`);
```

Explanation: Eight random scores are generated from 30 to 100. max/min find the extremes, reduce calculates the total, filter counts passes, and map formats each student's score.